* 0000800000001 * DFD

Cambridge Assessment International Education

Cambridge International AS & A Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

* 9 7 7 1 2 1 2 8 3 4 *

COMPUTER SCIENCE **9618/32**

Paper 3 Advanced Theory **October/November 2025**

1 hour 30 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].
- No marks will be awarded for using brand names of software packages or hardware.

This document has **16** pages. Any blank pages are indicated.

© UCLES 2025

1 The composite data type, `Car`, is defined in pseudocode as:

```
TYPE Car
   DECLARE RegNumber : STRING
   DECLARE Make : STRING
   DECLARE Model : STRING
   DECLARE BodyStyle : STRING
   DECLARE Colour : STRING
   DECLARE IntoStock : DATE
   DECLARE Price : REAL
ENDTYPE
```

(a) **(i)** Write the **pseudocode** statement to set up a variable for one record of the composite data type, `Car`.

....

.... [1]

(ii) Write the **pseudocode** statements to assign the following values to the variable set up in part **(a)(i)**:

- `"Blue"` to `Colour`
- `21/10/2025` to `IntoStock`

....

....

....

.... [2]

(b) The data type for `BodyStyle` is changed to an enumerated type, `Body`.

(i) Write the **pseudocode** statement for the type declaration of `Body` to hold the names of the available choices:

Convertible, Hatchback, Saloon, SUV

....

....

....

.... [2]

(ii) Write the new **pseudocode** statement required to update the declaration of `BodyStyle` in the definition of `Car`.

....

.... [1]

DO NOT WRITE IN THIS MARGIN

2 Numbers are stored in a computer system using binary floating-point representation with:

- 10 bits for the mantissa
- 6 bits for the exponent
- two's complement form for both the mantissa and the exponent.

(a) Calculate the denary value of the given normalised binary floating-point number.
Show your working.

Mantissa

0	1	1	1	1	0	0	1	0	1

Exponent

0	0	1	0	1	1

Working ....

....

....

....

....

Denary value ....
[3]

(b) Calculate the normalised binary floating-point representation of +26.6875 in this system.
Show your working.

Mantissa

Exponent

Working ....

....

....

....

....

....
[3]

DO NOT WRITE IN THIS MARGIN

3 (a) HTTP and IMAP are examples of protocols used in the Application Layer of the TCP/IP protocol suite.

State the purpose of the HTTP and IMAP protocols.

HTTP ....

....

IMAP ....

.... [2]

(b) Describe how files are shared using the BitTorrent protocol.

....

....

....

....

....

....

....

.... [4]

4 (a) Identify **one** benefit of circuit switching **and one** benefit of packet switching.

Circuit switching ....

....

Packet switching ....

.... [2]

(b) Identify **two** differences between circuit switching and packet switching.

1 ....

....

....

2 ....

....

.... [2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

5 (a) Describe how interrupt handling is used in low-level scheduling.

....

....

....

....

....

.... [2]

(b) In process management, a process can be in one of three process states: running, ready or blocked.

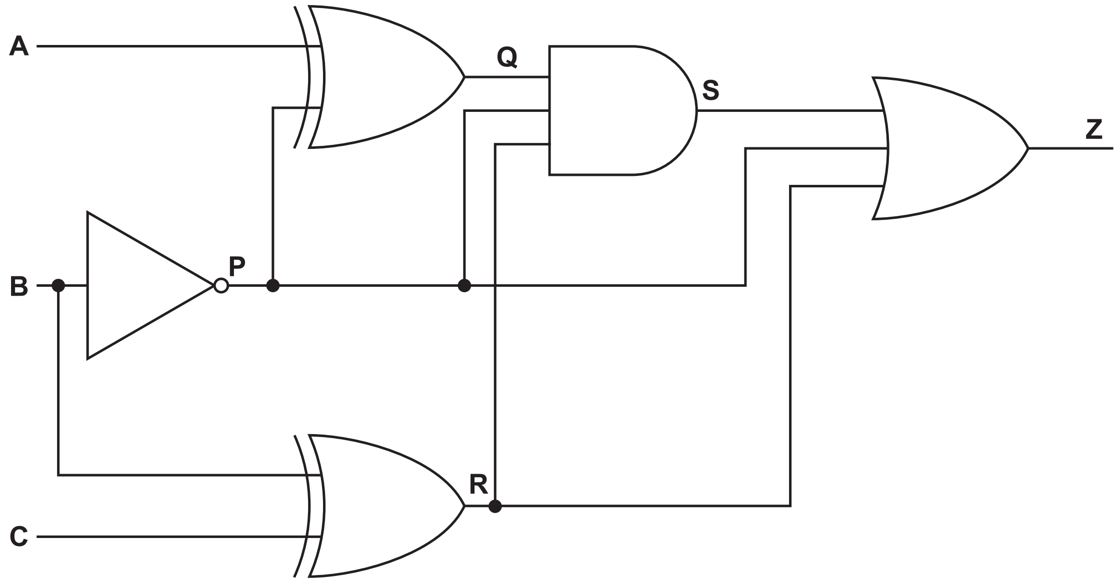
Complete the table to identify **one** reason why a process could be in each of the three states.

Process state	Reason
running	....
ready	....
blocked	....

[3]

DO NOT WRITE IN THIS MARGIN

6 (a) The diagram shows a logic circuit.



Complete the truth table for the given logic circuit.
Show your working.

			Working space				
A	**B**	**C**	**P**	**Q**	**R**	**S**	**Z**
0	0	0					
0	0	1					
0	1	0					
0	1	1					
1	0	0					
1	0	1					
1	1	0					
1	1	1					

[3]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(b) **(i)** Complete the Karnaugh map (K-map) for the Boolean expression:

$\mathbf{\overline{A}.\overline{B}.C + \overline{A}.B.C + A.\overline{B}.C + A.B.\overline{C}}$

A \ BC	**00**	**01**	**11**	**10**
0				
1				

[2]

(ii) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

(iii) Write the Boolean expression from your answer to part **b(ii)** as a simplified sum-of-products. Do **not** carry out any further simplification.

....

.... [2]

7 **(a)** Asymmetric encryption is a type of cryptography.

Identify **one** other type of cryptography.

....

.... [1]

(b) An organisation holds two asymmetric encryption keys, which they intend to use to receive secure transmissions.

Explain how the organisation makes use of the two keys to receive a secure transmission.

....

....

....

....

....

....

....

.... [4]

8 **(a)** State the purpose of the optimisation stage in the compilation of a program.

[1]

(b) Convert this Reverse Polish Notation (RPN) back to its original infix form:

`a b - c + c a - * d /`

[3]

(c) The RPN expression:

`c a / b d - * b +`

is to be evaluated, where:

`a = 3`, `b = 16`, `c = 9` and `d = 6`.

Show the changing contents of the stack as the RPN expression is evaluated.

[4]

DO NOT WRITE IN THIS MARGIN

9 (a) Deep Learning is a form of Machine Learning.

(i) State **one** example where Deep Learning is used.

....

.... [1]

(ii) Identify how Deep Learning can be made more effective.

....

.... [1]

(b) Describe the back propagation of errors method in Machine Learning.

....

....

....

....

....

....

....

.... [4]

10 An exception is an error that may cause a program to halt unexpectedly.

(a) Describe how program termination due to an exception can be avoided.

....

....

....

....

.... [2]

(b) Identify **two** possible causes of exceptions.

1 ....

....

2 ....

....

[2]

DO NOT WRITE IN THIS MARGIN

11 The table shows assembly language instructions for a processor that has one register, the Accumulator (ACC).

Label	Instruction		Explanation
	Opcode	Operand	
	LDM	#n	Load the number n to the ACC
	LDD	<address>	Load the contents of the location at the given address to ACC
	LDI	<address>	The address to be used is at the given address. Load the contents of this second address to the ACC.
	ADD	<address>	Add the contents of the given address to the ACC
	SUB	<address>	Subtract the contents of the given address from the ACC
	STO	<address>	Store the contents of the ACC at the given address
<label>:		<data>	Gives a symbolic address <label> to the memory location with the contents <data>
# denotes a denary number, e.g. #123 <label> can be used in place of <address>			

The current contents of memory are:

Address	Contents
150	26
300	86
420	150

DO NOT WRITE IN THIS MARGIN

Write **assembly language** code, using **only** the given instruction set to:

- store the contents of location 300 as labelled variable `A`
- store the contents of location 420 as labelled variable `B`
- add the value stored in the address contained in variable `B` to the value contained in variable `A`
- store the result in variable `Answer`.

Show the initialisation and values of the variables `A`, `B` and `Answer` in the table provided.

....

....

....

....

....

....

....

....

Label	**Content**

[7]

DO NOT WRITE IN THIS MARGIN

12 (a) A stack has been implemented using pseudocode to store a maximum of 100 string items using the global variables in the following table:

Identifier	Data type	Description	Initialisation value
`Base`	`INTEGER`	pointer for the bottom of the stack	`0`
`Top`	`INTEGER`	pointer for the top of the stack	`-1`
`StackArray`	`STRING`	1D array to implement the stack	`[0:99]`
`Max`	`INTEGER`	maximum number of items in the stack	`100`

The value of `Top` is incremented each time a data item is added to the stack and decremented each time a data item is removed. If the stack is full, an appropriate error message is output.

(i) Complete the **pseudocode** for the procedure to add a data item onto the stack.

```
PROCEDURE Push(..............................................................)
   IF Top < Max - 1 THEN

      Top ← .........................................................................

      ............................................................... ← NewData
   ELSE

      OUTPUT .......................................................................
   ENDIF
ENDPROCEDURE
```

[4]

(ii) Write **pseudocode** to input a new data item and add it to the stack using `Push()`.

....

....

....

.... [2]

(b) Explain the reasons why a stack is used when a recursive algorithm is executed.

....

....

....

....

....

.... [3]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

BLANK PAGE

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

BLANK PAGE

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

BLANK PAGE

BLANK PAGE

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.